JAVA MINI PROJECT DOCUMENTATION

Banking System Simulation (GUI + MySQL)

---

1. PROJECT OVERVIEW

The **Banking System Simulation** is a desktop-based application developed using **Core Java (Swing)** and **MySQL database (JDBC)**. The system replicates real-world banking operations such as user authentication, account handling, transactions, and audit tracking.

This application provides:

- Secure login and registration
- Real-time transaction processing
- Persistent data storage
- User-friendly graphical interface

All financial operations are recorded and stored in a relational database, ensuring **data integrity and auditability**.

---

2. OBJECTIVES AND KEY CONCEPTS COVERED

🎯 Objectives

- Develop a GUI-based banking application using Java Swing
- Implement secure authentication with validation
- Perform core banking operations (Deposit, Withdraw, Transfer)
- Store and manage data using MySQL
- Track transaction and login history
- Demonstrate real-world software design principles

□ Key Concepts Covered

Concept	Description
Object-Oriented Programming	Structured design using classes and methods
Encapsulation	Data and logic bundled within classes
Abstraction	Hiding database logic behind methods

Concept	Description
JDBC	Java Database Connectivity for MySQL
SQL (CRUD Operations)	Insert, Update, Select operations
Swing GUI	User interface components
Event Handling	ActionListener for user interaction
Collections (ArrayList)	Dynamic data handling
Exception Handling	try-catch for runtime errors
Data Validation	Input validation for secure processing

---

3. SYSTEM ARCHITECTURE

The system follows a **layered architecture**:

```
User Interface (Swing GUI)
        ↓
Application Logic (Java Classes)
        ↓
Database Layer (MySQL via JDBC)
```

- GUI handles user interaction
- Business logic processes operations
- Database stores persistent data

---

4. CLASSES AND THEIR RESPONSIBILITIES

Class Name	Role
MainApp	Entry point of the application
DBConnection	Manages database connectivity
LoginFrame	Handles login and registration
DashboardFrame	Handles banking operations

---

5. CLASS DESCRIPTION

5.1 DBConnection

Purpose:

Establishes connection between Java application and MySQL database.

Key Function:

- Loads JDBC driver
- Connects using DriverManager
- Returns Connection object

5.2 LoginFrame

Purpose:

Handles user authentication and registration.

Features:

- User Login
- New User Registration
- Input Validation
- Login History Tracking

5.3 DashboardFrame

Purpose:

Provides core banking functionalities after login.

Features:

- Deposit money
- Withdraw money
- Transfer funds
- Balance inquiry
- Transaction history (JTable UI)

---

6. CORE BANKING OPERATIONS

Deposit

- Adds amount to user balance
- Updates database
- Logs transaction

Withdraw

- Checks available balance
- Deducts amount
- Prevents overdraft

Transfer

- Transfers funds between users
- Updates both accounts
- Records sender and receiver transactions

---

7. DATABASE DESIGN

Database Name: `bank_system`

Table: users

Field	Type	Description
id	INT (PK)	Unique user ID
username	VARCHAR	Username
password	VARCHAR	Password

Table: accounts

Field	Type	Description
id	INT (PK)	Account ID
user_id	INT (FK)	User reference
balance	DOUBLE	Current balance

Table: transactions

Field	Type	Description
id	INT (PK)	Transaction ID
user_id	INT	User reference
type	VARCHAR	Transaction type
amount	DOUBLE	Amount
receiver	VARCHAR	Receiver username
date	TIMESTAMP	Transaction time

Table: login_history

Field	Type	Description
id	INT (PK)	Record ID
username	VARCHAR	Username
login_time	TIMESTAMP	Login timestamp

---

8. TRANSACTION HISTORY (JTABLE)

The system uses **JTable** to display transaction records in a structured format.

Columns:

- Transaction Type
- Amount
- Receiver
- Date

Advantages:

- Clear visualization
- Scrollable interface
- Professional UI

---

9. INPUT VALIDATION

Field	Validation Rule
Username	Cannot be empty
Password	Cannot be empty
Amount	Must be greater than zero
Transfer	Receiver must exist
Withdraw	Must not exceed balance

---

10. ERROR HANDLING

Scenario	Handling
Empty input	Error message displayed
Invalid login	"Invalid credentials"
Insufficient balance	Transaction blocked
Invalid receiver	Error message
Database error	Exception handling

---

11. FEATURES IMPLEMENTED

- GUI-based system (Swing)
- Secure Login & Registration
- Deposit, Withdraw, Transfer
- Transaction History (Table View)
- Login History Tracking
- Input Validation
- Persistent Storage (MySQL)

---

12. ADVANCED FEATURES

- Dual transaction recording (transfer)
- JTable for professional UI display
- Audit logging (transactions + login history)
- Multi-user support

---

13. FUTURE ENHANCEMENTS

- Password encryption (SHA-256)
- Transaction rollback (ACID properties)
- JavaFX modern UI
- REST API integration (Spring Boot)
- Mobile application extension

---

14. CONCLUSION

The Banking System Simulation successfully demonstrates **real-world financial operations** using Java and MySQL.

The project integrates:

- Object-Oriented Design
- GUI Development
- Database Management
- Secure Transaction Handling

It serves as a strong foundation for building enterprise-level banking applications.